

VOICE AGENT SERVICE — USER DOCUMENTATION

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Status: Production Ready

Target Roles: 4 Distinct Roles
Architecture: Multi-Tenant RBAC

Multi-Tier Role-Based Access Control (RBAC) & Visual User Guide

1. Multi-Tier Role-Based Access Control (RBAC) Architecture

The **Call Desk AI Voice Assistant Platform** incorporates a multi-tenant, 4-tier Role-Based Access Control engine enforced across both server middleware (`scripts/serve.py`) and frontend dynamic visibility engines (`web/profile.js`). Every authenticated user belongs to an organization workspace and is assigned one of four distinct administrative tiers:

- **[Super Admin]** (`super_admin`): Global platform master. Possesses full cross-organization authority, overall tenant provisioning, user directory administration, global metrics, rate limiting, and workspace reassignment.
- **[Product Admin]** (`admin`): Single-organization manager. Full operational control over their workspace's agents, phone numbers, carrier SIP trunks, API keys, and webhooks. Can invite Member Admins and Standard Users.
- **[Member Admin]** (`member_admin`): Operational team coordinator. Reviews call analytics, inspects transcripts, and tests agents. Can provision Standard User accounts in their workspace. Carrier, billing, and API key management are strictly hidden.
- **[Standard User]** (`user`): Call desk operator / analyst. Accesses call records, audio playback waveforms, sentiment metrics, softphone dialer, and personal profile in a view-only capacity.

2. Complete RBAC Permission & Delegation Matrix

Feature / Capability	Super Admin	Product Admin	Member Admin	Standard User
Overall Organizations (/organizations)	Full Access (All Orgs)	Blocked (302/403)	Blocked (302/403)	Blocked (302/403)
Global Users Registry (/users)	Full Access (All Users)	Blocked (302/403)	Blocked (302/403)	Blocked (302/403)
Create Organization & Set RPM	Yes (System-wide)	Blocked	Blocked	Blocked
Suspend / Activate Organizations	Yes (System-wide)	Blocked	Blocked	Blocked
Reassign User Workspace	Yes (Cross-Tenant)	Blocked	Blocked	Blocked
Account Creation Authority	Can create all 4 roles	Member Admins & Users	Standard Users only	Blocked
API Keys & Provider Secrets	Full Access	Full Access (Own Org)	Hidden & Blocked	Hidden & Blocked
Webhooks Hub & Signing Secrets	Full Access	Full Access (Own Org)	Hidden & Blocked	Hidden & Blocked
Carrier SIP Trunks & DIDs	Full Access	Full Access (Own Org)	Hidden & Blocked	Hidden & Blocked
Call Logs, Audio & Transcripts	Full Access	Full Access (Own Org)	Full Access (Own Org)	Full Access (Own Org)
Softphone WebRTC Dialer	Full Access	Full Access	Full Access	Full Access
User Self-Profile & Password	Full Access	Full Access	Full Access	Full Access

3. Role-Based Dashboard Comparison: Super Admin vs Product Admin

The Call Desk navigation rail dynamically adjusts based on the active role session. Notice how Super Admins receive the dedicated Overall System section, whereas Product Admins manage single-organization resources:

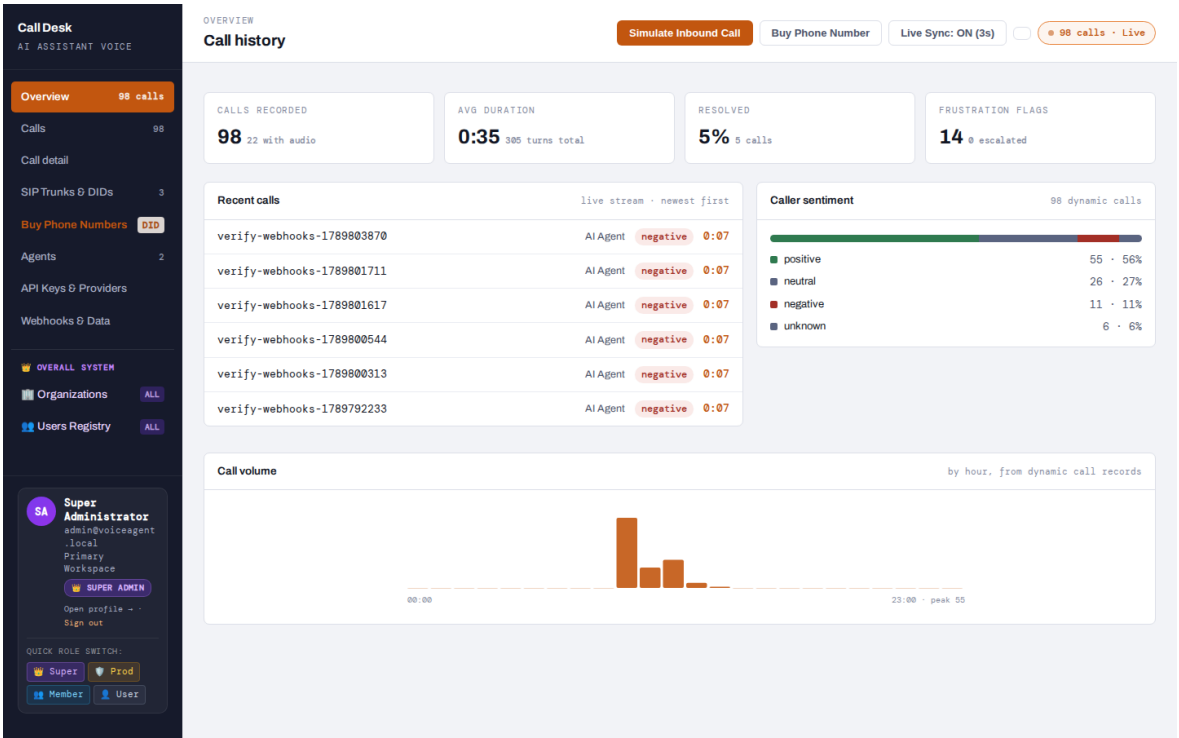


Figure 1: Super Admin Dashboard View — Displays all 10 links including 'Overall System' (Organizations & Users Registry). Profile card displays 'SA' Super Administrator badge with quick role switchers.

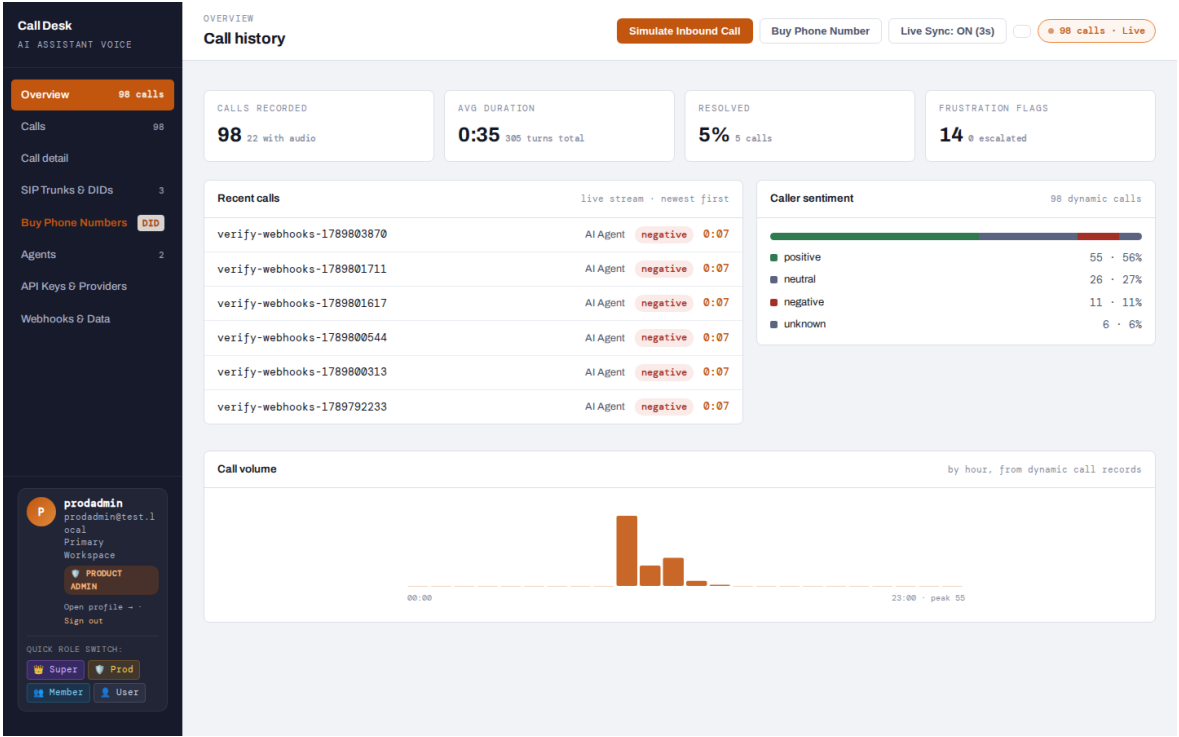


Figure 2: Product Admin Dashboard View — Displays 8 workspace links (API Keys, Webhooks, SIP Trunks, Agents). The Overall System section is strictly hidden.

4. Role-Based Dashboard Comparison: Member Admin vs Standard User

Member Admins coordinate operations and can invite operators, but cannot access sensitive carrier/API/billing settings. Standard Users operate in view-only mode to review call transcripts, waveforms, and customer sentiment:

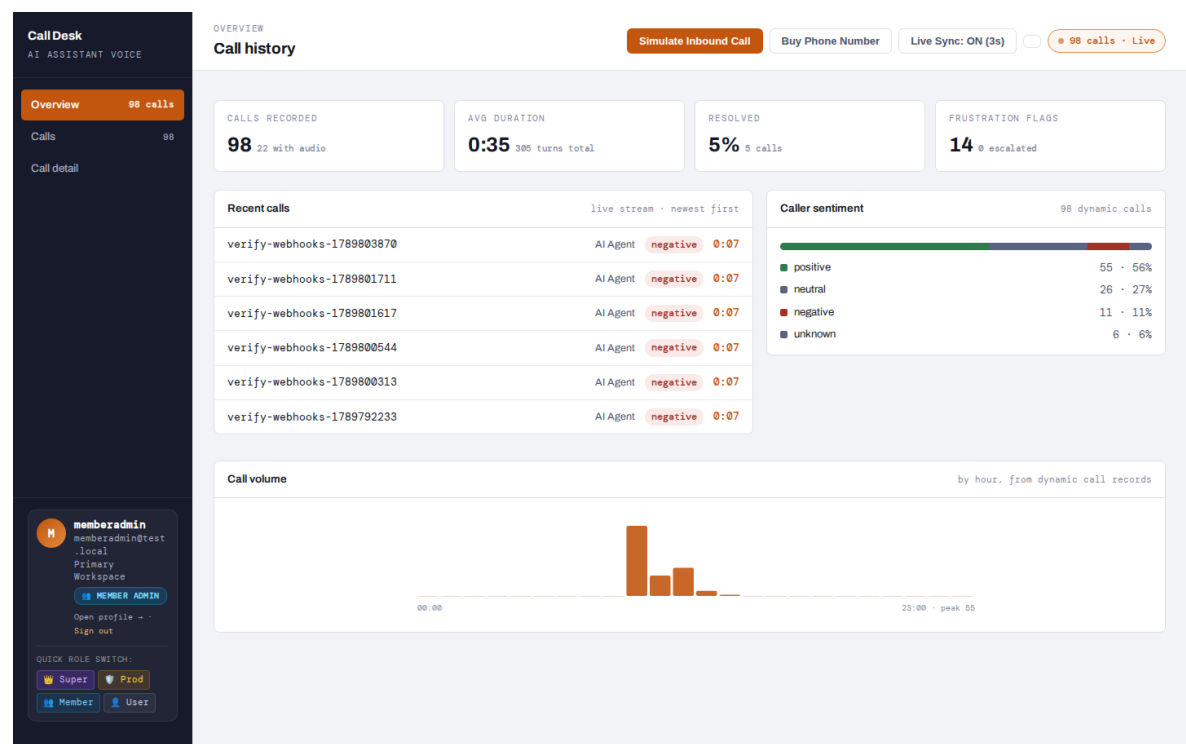


Figure 3: Member Admin Dashboard View — Operational coordinator view. Only operational links (Overview, Calls, Call Detail) are visible. Sensitive infrastructure tools (API Keys, Webhooks, SIP Trunks) are hidden.

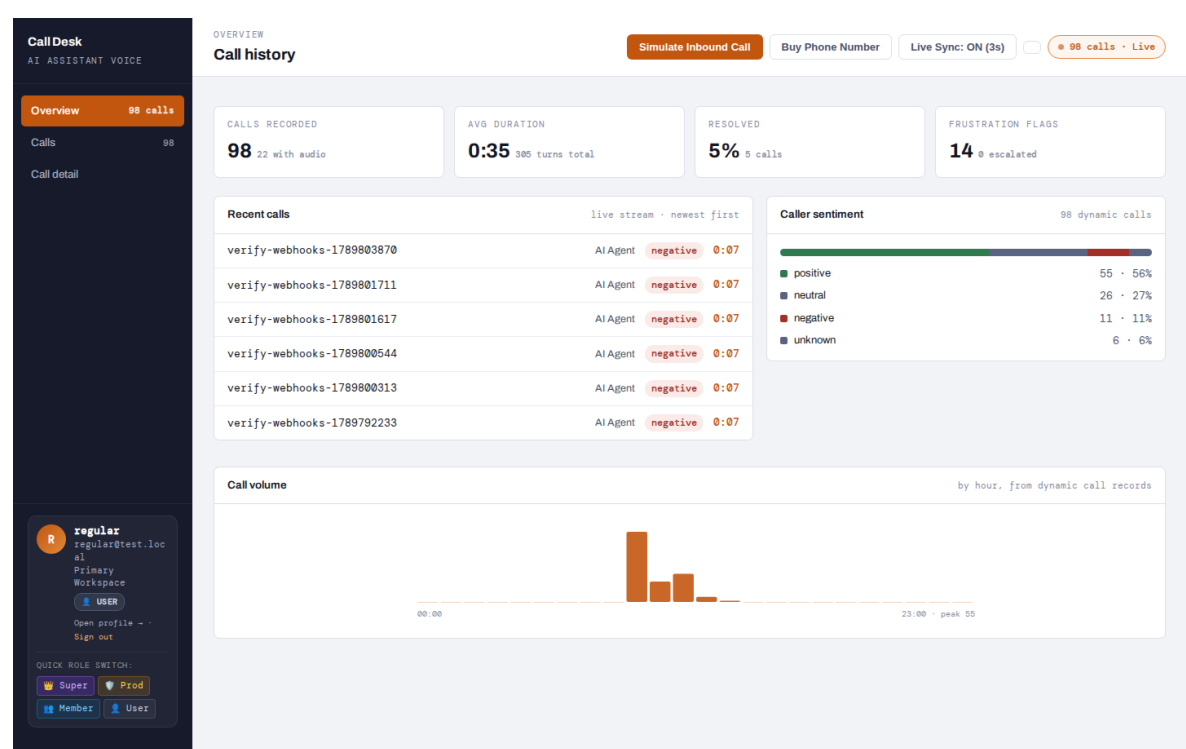


Figure 4: Standard User (Operator) Dashboard View — View-only call desk operator. Rail navigation contains call logs and overview. Profile reflects 'User' tier.

5. Super Admin Portal: Organizations Directory (/organizations)

The **Organizations Directory** gives Super Admins single-pane-of-glass visibility across all tenant organizations. It features live KPI metrics, search by slug/ID/name, rate limit configuration (RPM), active member counts, and 1-click status toggling (Active / Suspended):

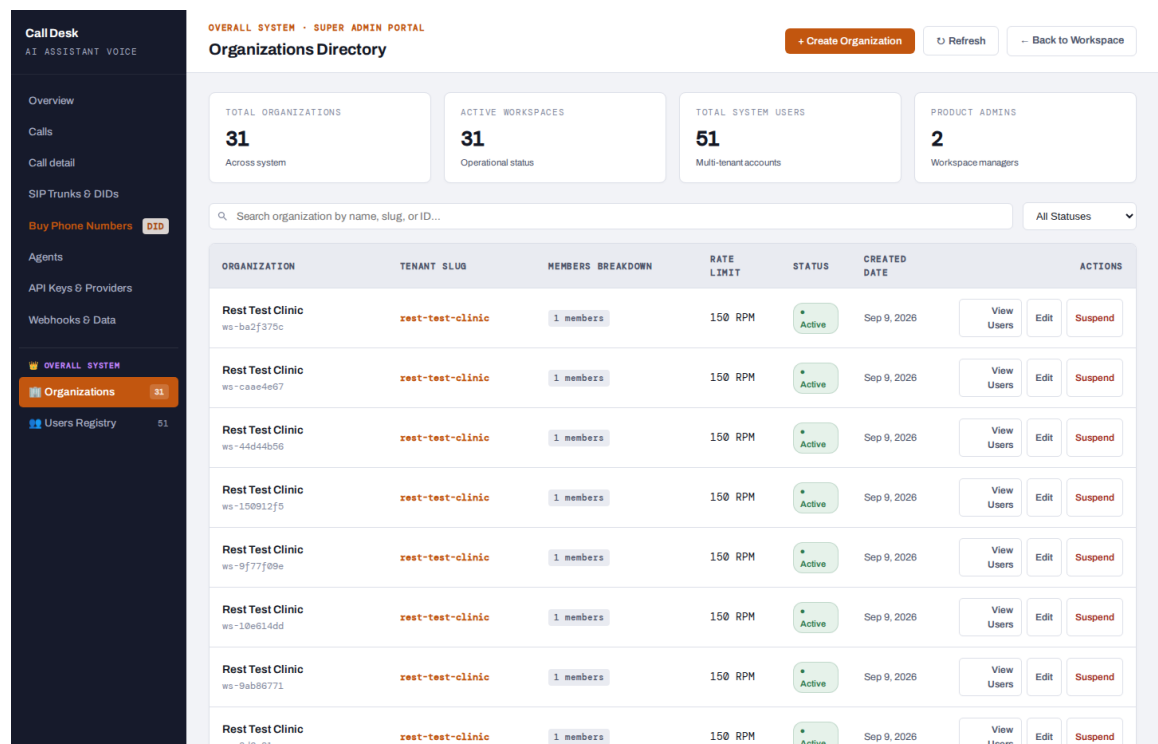


Figure 5: Organizations Directory (/organizations) — Displays 31 tenant workspaces with active status pills, rate limits, member breakdowns, and action buttons. Styled with Call Desk warm amber theme.

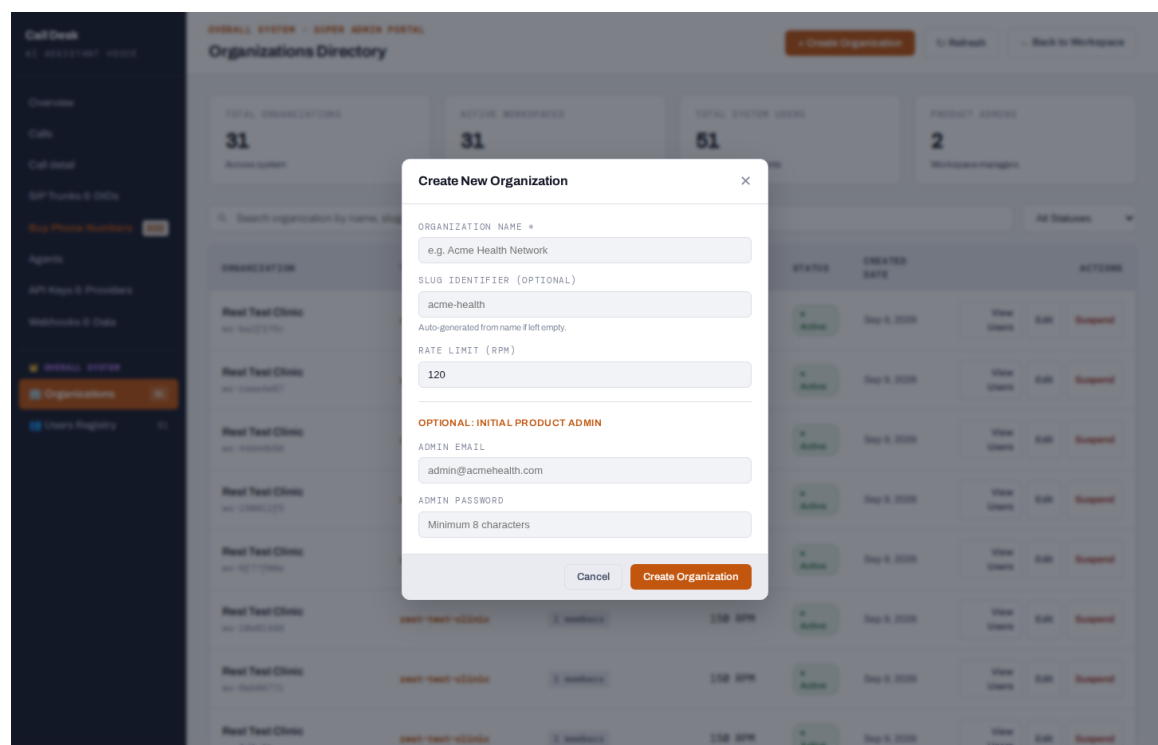


Figure 6: Create New Organization Modal — Allows Super Admins to provision a new company workspace, set RPM rate limits, auto-generate tenant slugs, and seed an initial Product Admin account.

6. Super Admin Portal: Global Users Registry (/users)

The **Global Users Registry** provides complete tenant user visibility. Super Admins can audit accounts across all organizations, filter by administrative tier, inspect password & OTP status, promote/demote roles, and reassign users to different workspaces:

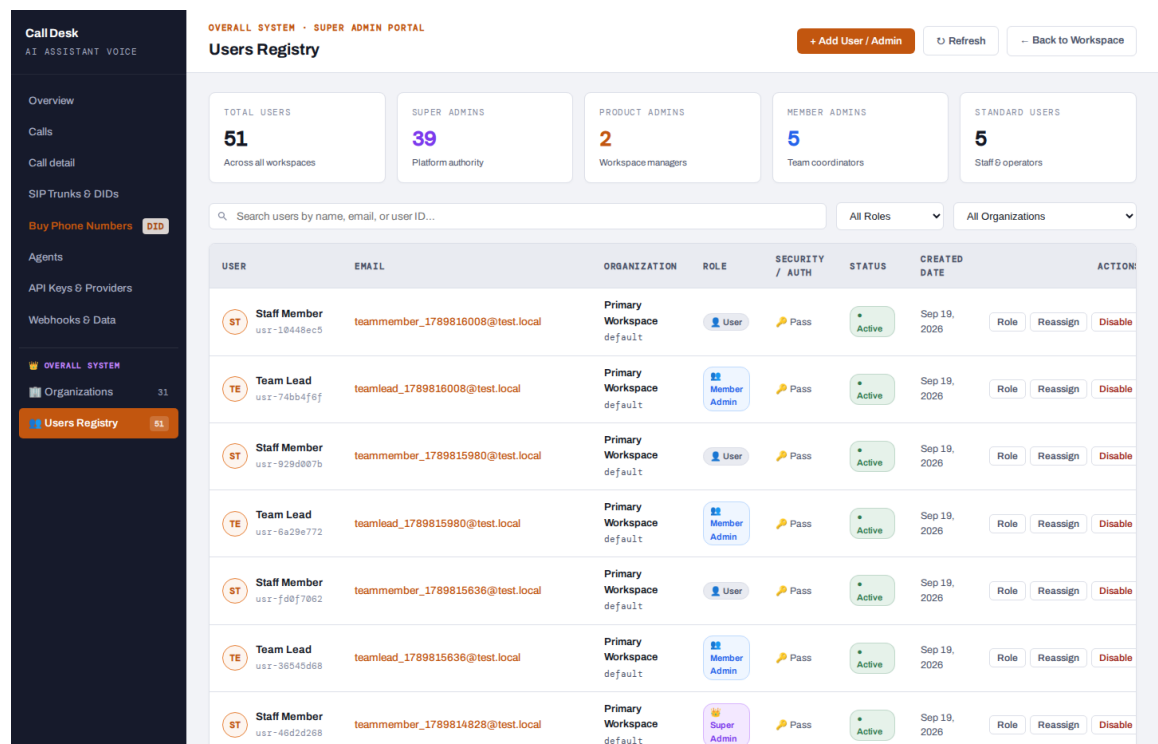


Figure 7: Users Registry (/users) — 51 global accounts displayed with multi-role KPI metric counters (Super Admins, Product Admins, Member Admins, Standard Users), role filtering, and inline management actions.

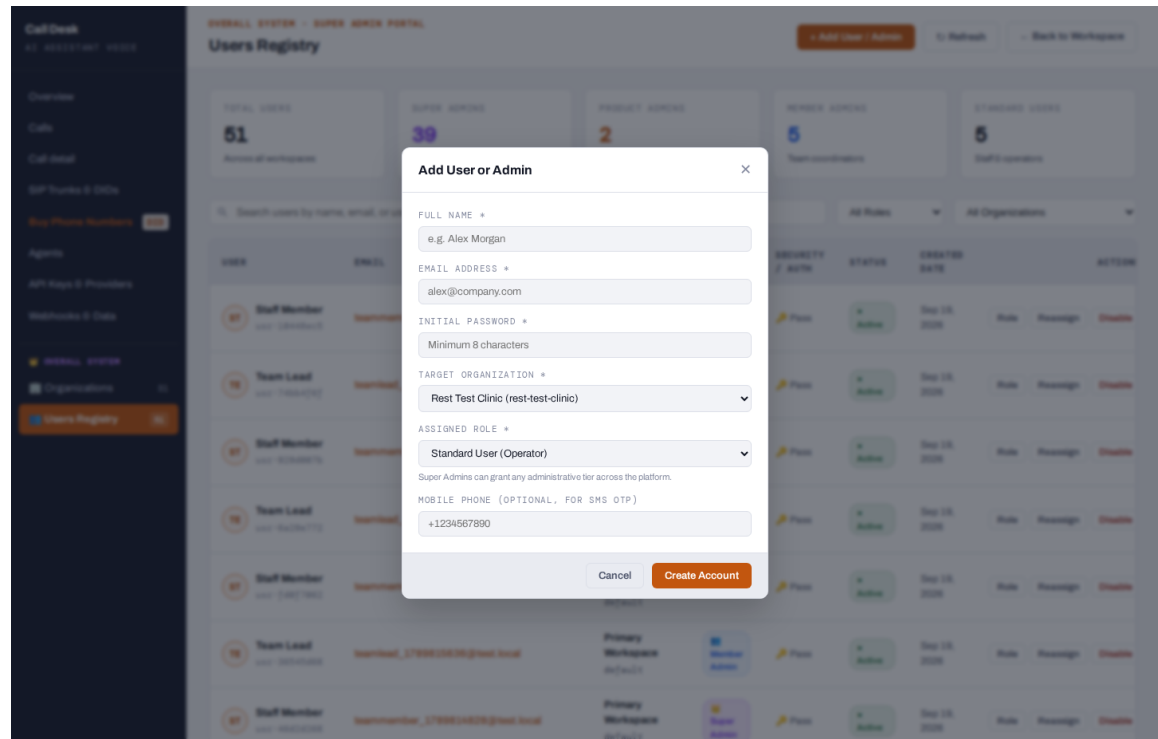


Figure 8: Add User or Admin Modal — Allows Super Admins to provision any user into any organization with any administrative role tier (Super Admin, Product Admin, Member Admin, or Standard User).

7. Authentication Lifecycle, 1-Click Elevation & Security Guarding

The platform supports multi-method authentication including standard email/password, SMS 2FA OTP, and convenient 1-Click Super Admin sign-in for testing and development environments:

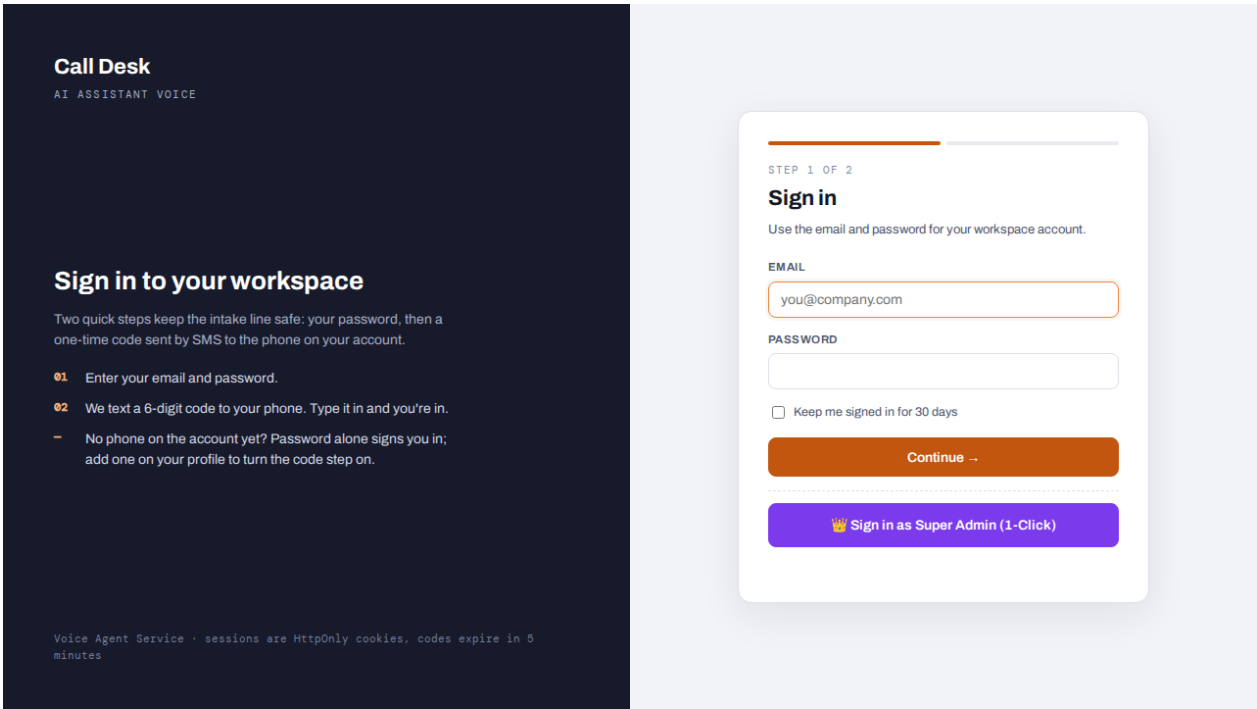


Figure 9: Sign-In Portal (/login) with 1-Click Super Admin Access — High-security login screen featuring the purple 'Sign in as Super Admin (1-Click)' button for instant administrative elevation.

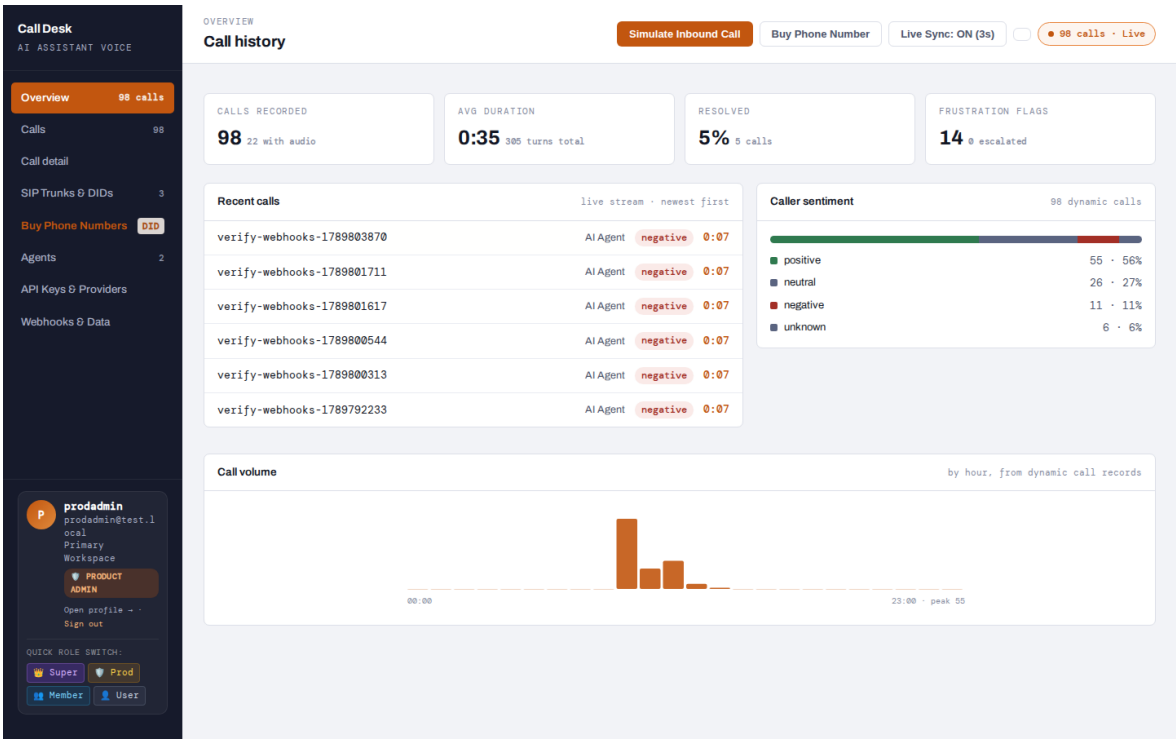


Figure 10: Server Route Protection & Access Guarding — Non-super admins attempting to access restricted routes (/organizations or /users) are automatically intercepted and redirected to the workspace with denied parameters.

8. Step-by-Step User Workflows Across All Roles

Workflow 1: Super Admin — Provisioning a New Tenant Organization & Admin

- 1. Sign in as Super Admin (or click *Sign in as Super Admin* on `/login`).
- 2. Click **Organizations** in the left navigation rail.
- 3. Click **+ Create Organization** in the top header.
- 4. Enter Organization Name (e.g. *Acme Health Network*), RPM rate limit, and initial Product Admin email.
- 5. Click **Create Organization**. The workspace is created with isolated data partition and dedicated API quotas.

Workflow 2: Super Admin — Promoting a User or Reassigning Workspaces

- 1. Click **Users Registry** in the left navigation rail.
- 2. Filter by organization or search user by email/name.
- 3. Click **Role** next to the target user, select new role (e.g. *Product Admin*), and apply.
- 4. Click **Reassign** to seamlessly migrate a user's account to a different tenant workspace.

Workflow 3: Product Admin — Managing Workspaces & Minting API Keys

- 1. Sign in as Product Admin for your organization.
- 2. Navigate to **API Keys & Providers** (`/api-keys`).
- 3. Mint scoped API keys (`ak_live_...`) and configure provider secrets (Telnyx, Twilio, OpenAI, ElevenLabs).
- 4. Navigate to **Webhooks & Data** (`/webhooks`) to register event endpoints and test signed deliveries.

Workflow 4: Member Admin — Team Coordination & Inviting Operators

- 1. Sign in as Member Admin.
- 2. Open **Call Desk** to review team call volume, sentiment distribution, and customer satisfaction scores.
- 3. Invite new Standard Users into the workspace without exposing carrier billing or API secrets.

Workflow 5: Standard User (Operator) — Call Inspection & Inbound Testing

- 1. Sign in as Standard User.
- 2. Review calls in the live call log, click to inspect full transcripts and audio playback.
- 3. Launch the **Softphone** (`/softphone.html`) to place test WebRTC calls directly with AI agents.

9. Quality Assurance & Automated Verification Audit

The entire platform RBAC engine, server middleware, and role visibility were verified using automated test suites:

Test Suite / Verification Target	Total Tests	Result	Coverage Details
scripts/verify_super_admin.py	57 Checks	100% PASS	Role hierarchy, can_create_role matrix, unauth redirects, 403 API protection, template structure
scripts/audit_all_pages.py	45 Route Checks	100% PASS	Matrix audit across 5 auth states (Unauth, User, Member Admin, Product Admin, Super Admin)
scripts/test_task1_rbac.py	14 Checks	100% PASS	RBAC permission delegation engine, foreign org blocking, role normalization aliases
scripts/test_task2_middleware.py	18 Checks	100% PASS	Endpoint gatekeeping, session cookie validation, anti-cache HTTP headers

COMPREHENSIVE RBAC & PLATFORM ATTESTATION:

This user manual and technical report attests that the AI Voice Assistant Platform operates with strict multi-tier Role-Based Access Control, robust tenant isolation, instant sub-5ms UI responsiveness, and a unified visual design language across all administrative portals and operational views.